BMS PLAYBOOK

Bakery Management System | Client-aligned MVP specification

An internal, responsive and installable web app for replacing paper-based bakery records. Prepared as the development brief for Antigravity.

Project	Agreed direction
Budget	NGN 250,000 total, including initial domain and deployment
Devices	Owner uses laptop/desktop; salespeople and distributors use mobile phones
Users	Owner, salespeople, distributors
Approach	Manual input, automatic calculations, tracking and reports; no payment gateway
Status	Prototype first, client review, then implementation

1. What the client actually wants

A simple digital register for daily bakery activity. Staff input sales, production, raw-material movements, distributor activity and customer payments. The app consolidates records on the owner dashboard, tracks balances and stock, and produces downloadable Excel reports and printable transaction documents. It is not an online shop or a full ERP.

2. MVP modules and functionality

Module	Included functionality
Login and roles	Owner, salesperson and distributor accounts; role-limited access. Owner manages staff.
Products	Owner creates/edit bakery products, categories, unit prices and active status. No hardcoded product list.
Sales	Agent selects products and quantities, records buyer and amount paid; app calculates total and outstanding balance. Cash/transfer/POS entered manually.
Customers and credit	Customer list, purchase history, part payments, balances and later repayment against the original sale. Walk-in option.
Production	Staff/owner enters quantities produced and damaged; usable quantity updates finished-goods stock.
Raw materials	Record ingredient stock received, used and wasted; show balances. Manual inputs, no recipe engine.
Distribution	Assign stock to distributor; log acknowledgement, sold/delivered quantities, returns and reconciliation. Transfer alone is not a sale.
Receipts / invoices	Generate clear sale document with total, paid and balance; print-friendly 58/80 mm layout, PDF/browser printing. Direct Bluetooth only after printer test.
Owner dashboard	Summary of sales, amounts collected, debts, finished stock, material balances, production and distributor activity. Basic date filters.
Excel	Export .xlsx sales, payments, customers, production, stock and outstanding balances; controlled import of initial products/customers/opening stock.
Offline PWA	Installed/cached app supports essential data entry offline, queues locally and syncs on reconnect; visible sync status.

3. Core workflows

Sale and partial payment

Agent selects customer (or walk-in) and products, enters quantity and recorded payment. Example: sale NGN 50,000; paid NGN 30,000; outstanding NGN 20,000. Receipt shows all three. When the remaining NGN 20,000 arrives, add a payment to the existing sale; do not create another product sale. Stock is deducted once per accepted sale.

Production and materials

Record produced and damaged finished items; add net usable units to finished stock. Separately record raw-material receipts, use and waste. Do not auto-deduct ingredients through a recipe system in this MVP.

Distributor

Owner assigns products, distributor confirms receipt and records delivery/sales and returns. Track allocated stock separately; allocation is not revenue. The owner views reconciliation.

Reports

Owner filters by period, views operational totals and exports an Excel workbook. Excel is an output/import format, not the live database. Live Google Sheets synchronisation is deferred unless separately agreed.

4. Permissions

Action	Owner	Salesperson	Distributor
Overall reporting / staff / price settings	Yes	No	No
Record sales and payments	Yes	Yes	Only if authorised
Production / raw-material entries	Yes or assigned staff	Only if assigned	No by default
Assigned stock and activity	All	Own if applicable	Own
Print relevant receipts	Yes	Yes	If authorised
Export full reports	Yes	No	No

5. Responsive product experience

- Laptop: calm desktop workspace with side navigation, useful summary cards, filterable records, legible data tables and straightforward forms.
- Mobile: large touch targets, short entry forms, compact cards and fast access to New Sale, assigned distribution tasks, receipts and sync status.
- One web app with role-specific screens; installable PWA rather than separate native apps.
- Visual direction: warm off-white, white surfaces, dark charcoal, restrained bakery-toned accent, Inter and Lucide; follow actual client branding when supplied. Avoid visual clutter.

6. Technology and architecture

Layer	Recommended choice
UI	React + Vite + TypeScript + Tailwind CSS + shadcn/ui + Lucide
Forms and server data	React Hook Form + Zod + TanStack Query
Backend	Supabase Auth + PostgreSQL + Row Level Security; secure database functions for critical writes
Offline	PWA service worker + IndexedDB/Dexie pending operations and cache
Reports	ExcelJS or equivalent maintained .xlsx library; print CSS/PDF
Deployment	Commercial-use-permitted static hosting such as Cloudflare Pages; custom domain and HTTPS

Supabase is the authoritative database. Google Sheets, if ever added, is a reporting destination, not a transactional database. Free service tiers require quota monitoring, backup arrangements and a future upgrade agreement. Do not promise permanent zero hosting cost.

7. Data entities (lean model)

Staff profiles/roles; products; customers; sales; sale items; payments; production entries; raw materials; raw-material movements; finished-stock movements; distributor transfers and their items; audit log. Include unique transaction IDs and timestamps. Add only supporting tables that are needed to implement these workflows safely.

8. Offline and simultaneous-use rules

- Save offline operations to IndexedDB with unique IDs, queue state and retry on reconnect/app resume. A service worker alone is not a transaction sync engine.
- Show Saved locally, Pending, Synced, Failed or Needs review. Never announce that a local operation is globally confirmed.
- Server processes repeated submissions idempotently; use atomic database operations for balances and stock.
- Agent A and Agent B can work independently; the owner sees consolidated updates after each device syncs. Disconnected devices cannot see each other in real time.
- If disconnected agents oversell shared inventory, retain records and flag the discrepancy for owner reconciliation; never silently overwrite or drop them.
- Offline use must be tested on actual target phones after the app has been installed and initial data cached.

9. Printing and reports

- Build a branded receipt/invoice preview with items, quantities, prices, total, amount received, balance, reference and date.
- Provide browser/PDF printing and 58mm/80mm print styles. Obtain the actual Bluetooth printer make/model and test with the intended phone/browser before committing to direct printing.
- Allow owner to export accurate .xlsx reports. Provide validated templates for initial products, customers and stock only; show import preview/errors before committing changes.

10. Do and do not

DO	DO NOT
Keep everyday input simple and role-specific.	Add e-commerce, checkout, gateway, AI or complex accounting.
Enforce permissions in backend and keep an audit trail.	Trust hidden frontend controls as security.
Use safe money calculations and append payment history.	Delete/rewrite sales to correct a payment.
Validate stock, sync and real-device PWA behaviour.	Assume offline devices have live shared inventory.
Test the actual thermal printer.	Promise universal Bluetooth/browser support.
Document backups, client ownership and costs.	Assume free hosting is unconditional forever.

11. Explicit exclusions and change control

No public storefront, online orders, payment gateway, automated bank reconciliation, payroll, multi-branch support, AI, advanced recipes/costing, GPS fleet tracking, WhatsApp API, native mobile apps or complex supplier procurement. Scheduled external messaging and live Google Sheets sync are enhancements, not default deliverables. Any newly requested feature requires a separate scope discussion.

12. Delivery stages and acceptance

Stage	Outcome
1. Prototype	Realistic demo flows for owner desktop and staff mobile: dashboard, sale/partial payment, customer, production, materials, distributor, receipt and report previews. Confirm workflows with client.
2. Core build	Authentication and permissions, products/customers, sales/payments, stock, production, materials and distribution connected to database.
3. Offline and outputs	PWA, local queue and reconciliation, Excel exports/import templates and receipt print testing.
4. Verification and launch	Test on laptop and mobile; multi-user/duplicate sync/partial payments/stock and roles; deploy domain, document backup, and hand over basic user guide.

Acceptance examples: A sale of NGN 50,000 with NGN 30,000 received leaves NGN 20,000 outstanding; a later payment settles that same sale; production increases usable stock; sale reduces stock once; distributor allocation does not count as a sale; queued offline entries sync without duplicate transactions; owner exports correct Excel reports and can print through the agreed tested workflow.

13. Client confirmations before final implementation

- Actual bakery name, logo, colours, product catalogue, prices and measurement units.
- Whether distributors sell independently or only deliver, including the treatment of returns and customer collections.
- Who records production and raw-material movement; assigned stock rules for agents.
- Exact Bluetooth printer model and target Android/iOS devices.
- Preferred report timing and whether she needs only in-app/downloadable reports or later scheduled notifications.
- Domain name, ownership, renewal responsibilities and any paid service/backup requirements.

14. Instructions for Antigravity

Read this playbook first. Do not expand the feature scope. Begin by proposing the responsive route map, lean database design, role matrix, offline queue rules and prototype screen flow. Flag ambiguous bakery business rules and printer constraints. Build incrementally, test each workflow and maintain concise README, architecture, database, offline-sync, deployment and user-guide documentation. The first deliverable is a reviewable prototype, not an uncontrolled full-system implementation.